BEHZAD SHAYEGH TheShayegh.github.io the.shayegh@gmail.com

Summary

Machine learning researcher and engineer with 7 years of R&D experience across academia and industry. Specialized in foundation models, agentic workflows, and large-scale AI evaluation methodologies. Proven track record of taking ML systems from foundational algorithmic design to production deployment. Experienced in cross-boundary collaboration to integrate complex LLM architectures into user-facing products. Published first-authored research at top-tier venues (NeurIPS, ICLR, ACL, AAAI).

Research Experience

Research Engineer at *RBC Borealis* *Since Jan. 2026*
– Agentic AI Research Group

- Designing and developing collaborative writing capabilities and evaluation for LLMs, composing long-form text by actively eliciting user preferences and domain knowledge efficiently.
- Designing and developing auto style-guide evaluation and generation, controlling the style of LLM outputs by automatically writing style guides as skills.
- Designed and developed deep market research and signal intelligence systems, providing critical strategic insights continuously utilized by RBC's C-suite and senior management teams.
- 🏆 Recognized with two internal awards for cross-team enablement and knowledge sharing, including representing RBC at external quantum hackathons and driving GenAI education initiatives.

Research Intern *Sep. 2025 – Jan. 2026*

- Formalized and evaluated context-certainty obedience of LLMs, and identified systematic limitations.
- Improved context-certainty obedience by 25% across eight diverse models (Gemma, Qwen, Llama; 1–72B) by an interaction strategy comprising prior reminders, certainty recalibration, and context simplification.

Affiliate Researcher *Sep. 2023 – Aug. 2024*

- Addressed over-smoothing in multi-teacher knowledge distillation by proposing ensemble distillation.

Research Intern at *Google* *Apr. 2025 – Jun. 2025*
– Machine Translation Research Group

- Uncovered and reduced adequacy-over-fluency bias in translation evaluation and meta-evaluation.
- Developed AdequacyX and FluencyX as specialized MetricX variants for adequacy and fluency evaluation, achieving 80% soft pairwise accuracy on their respective objectives.

Research Assistant at *University of Alberta* *2022 – 2025*
– NLP Research Group

- Achieved 3% improvement in 3 unsupervised parsing tasks across 3 languages by developing post-hoc aggregation methods and dynamic programming algorithms for ensemble building.
- Replicated 15 unsupervised parsers with different architectures: transformers, tensor decompositions, etc.
- Mentored an undergraduate student from project launch to AAAI co-authorship.

Research Engineer at *Divar* (classified ads platform with 140M annual ads) *Feb. 2022 – Jul. 2022*
– Used-Car Marketplace Team

- Curated a high-quality 1M-sample training dataset for car-price estimation using Spark: Detected outliers and interpolated missing entries.
- Designed an evaluation metric for car-price interval prediction that incorporates interval length as a proxy for uncertainty, and normalizes it by the real-world standard deviation of comparable data.
- Reduced car-price estimation uncertainty by 40%: Replaced the legacy over-complicated XGBoost model with an ensemble of quantile regressions using Scikit-learn.

- Improved car-price estimation MAPE by 4%: Identified and leveraged previously unrecognized and unaccounted categorical features using Gaussian Mixture Models and goodness-of-fit tests.

Research Intern (part time) at *Ashianeh-Sepehr-Kherad* *Jun. 2019 – Jun. 2021*
– Real Estate Machine Learning Group

- Built a 700K-sample dataset for address geolocation: Weighted data to match the population distribution.
- 81% accuracy in geolocating addresses from informal text under data scarcity by developing a hierarchical kernel-based Bayesian model.
- 82x speed-up and +4% accuracy for address geolocation: Developed an unsupervised statistical preprocessing model, reducing token variety from 85K to 11K.
- Engineered 400+ features for real-estate price estimation: Developed a feature-extraction pipeline using OpenStreetMap, PostgreSQL, QGIS, Python, and C++; Curated a 1M-sample training dataset accordingly.

Publications

Selected Publications

Behzad Shayegh, Mohamed Osama Ahmed, Fred Tung, Leo Feng. 2026. Can LLMs take retrieved information with a grain of salt?. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Behzad Shayegh, Yuqiao Wen, Lili Mou. 2024. Tree-averaging algorithms for ensemble-based unsupervised discontinuous constituency parsing. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL)*. Oral (top 2.5% of submissions)
└ Won ACL area chair's award, recognizing the best paper in its track.

More Publications

Behzad Shayegh, Niloofar Kazemi. 2026. Mind which bird you favour: Parameterizing adequacy–fluency balance in meta-evaluation of machine translation. In *Proceedings of the Conference on Machine Translation (WMT)*. Independent research

Behzad Shayegh, Jan-Thorsten Peter, David Vilar, Tobias Domhan, Juraj Juraska, Markus Freitag, Lili Mou. 2025. Feeding two birds or favoring one? Adequacy–fluency tradeoffs in evaluation and meta-evaluation of machine translation. In *Proceedings of the Conference on Machine Translation (WMT)*.

Yuqiao Wen, **Behzad Shayegh**, Chenyang Huang, Yanshuai Cao, Lili Mou. 2025. EBBS: An ensemble with bi-level beam search for zero-shot machine translation. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*. Also presented at *NGSM@ICML'24* and *SPIGM@ICML'24* workshops.

Behzad Shayegh, Hobie H.-B. Li, Xiaodan Zhu, Jackie C.K. Cheung, Lili Mou. 2025. Error diversity matters: An error-resistant ensemble method for unsupervised dependency parsing. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*.

Behzad Shayegh, Yanshuai Cao, Xiaodan Zhu, Jackie C.K. Cheung, Lili Mou. 2024. Ensemble distillation for unsupervised constituency parsing. In *International Conference on Learning Representations (ICLR)*.

Education

M.Sc. at *University of Alberta* *2022 – 2025*
– Major in computer science (GPA $\frac{4.0}{4.0}$)
– Thesis (2025). Ensemble methods for unsupervised syntactic parsing.
└ Runner-up for the departmental best MSc thesis award.

B.Sc. at *University of Tehran* *2017 – 2022*
– Major in computer engineering (GPA $\frac{18.2}{20}$) and minor in computer science (GPA $\frac{17.2}{20}$)
– Thesis (2021). Geolocating addresses from informal Persian texts using hierarchical kernel-based Bayesian models.

Teaching and Service

Mentorship

Research Mentor at *RBC Borealis*
– Let's solve it (LSi) program *(Fall 2026)*

Research Mentor at *NLP Group, University of Alberta*
– First research project of senior bachelor Hobie H.-B. Li, resulting in collaborative work published at AAAI *(Fall 2023)*

Research Mentor at *Computing Research Association (CRA)*
– Undergraduate research to Ph.D. (UR2PhD) program *(Fall 2023)*

Mentor at *ACM Student Chapter, University of Tehran*
– Summer of code: intro to artificial intelligence *(Summer 2019)*

Lectures and Course Development

Guest Lecturer for Machine Learning at *University of Alberta*
– Max a posteriori estimation *(February 2025)*
– Maximum likelihood estimation *(September 2024)*
– Gradient descent *(September 2024)*

Guest Lecturer for Discrete Mathematics at *University of Tehran*
– Invariance principle *(September 2019)*

Module Architect at *University of Tehran*
– How to write proofs in discrete mathematics

Content Developer at *Alberta Machine Intelligence Institute (AMII)*
– Explainability and transparency in machine learning

Content Developer at *University of Tehran*
– Introduction to discrete mathematics
– Common mistakes in discrete mathematics
– Applied probability and statistics in R
– Basic C++ exercises

Content Developer at *Tehran Polytechnic*
– Python tutorial

Teaching Assistantship — 18 offerings, 9 courses

Lead TA at *University of Tehran*
– Discrete mathematics $\times 3$
– Probability and statistics

TA at *University of Alberta*
– Machine learning $\times 2$
– Intro to the foundations of computation II $\times 2$

Volunteer TA at *University of Tehran*
– Natural language processing $\times 2$
– Artificial intelligence
– Discrete mathematics $\times 2$
– Probability and statistics
– Data structures and algorithms
– Intro to computing systems and programming
– Computer programming $\times 2$

Peer Review

- **Reviewer** for the 40th Annual Conference on Neural Information Processing Systems (NeurIPS), 2026.
- **Emergency Reviewer** for ACL Rolling Review (ARR), February cycle, 2025.
- **Reviewer** for the 13th International Conference on Learning Representations (ICLR), 2025.

Skills

Programming	Python (PyTorch, Hugging Face, vLLM, SciPy, NumPy, Pandas), C/C++, SQL, R, LaTeX
ML Methods	Unsupervised/self-supervised learning, contrastive learning, zero-/few-shot transfer, knowledge distillation, ensemble methods, gradient boosting (XGBoost), decoding algorithms (beam search, dynamic programming), synthetic data, explainability
Deep Learning	Transformers, encoder–decoder models, RNNs/LSTMs, CNNs, GNNs, GANs
LLMs & NLP	Agentic workflows, MCP, proprietary and open-weight LLMs, prompt engineering, LLM evaluation and meta-evaluation, RAG, machine translation, style transfer
Statistics	Estimation and Bayesian inference (MLE, MAP, EM, MCMC, variational inference, Gaussian processes), nonparametric methods, hypothesis testing (bootstrap, permutation, goodness-of-fit, multiple-comparison correction), uncertainty quantification
Infra & Tools	Linux/Bash, Docker, Slurm, CUDA, Apache Spark, PostgreSQL